

Resilience and Unconscious Coping

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RESILIENCE AND UNCONSCIOUS COPING

The mechanisms of defense serve the purpose of keeping off dangers. It cannot be disputed that they are successful in this; and it is doubtful whether the ego could do without them altogether during its development.

—SIGMUND FREUD

ONCE UPON A TIME at an amusement park in Florida, I watched some passengers (including my grandson) on a loop-the-loop roller-coaster. They gathered speed, swept up the curve, and hung suspended upside down at the top, waving their arms with excitement. I could see that for them the experience was one of ecstasy, exhilaration, and release. But it seemed to me that their elation, like the Ode to Joy of the angry and depressed Beethoven, reflected serious denial. For me, there would be nothing even remotely pleasurable in an experience like that. Just thinking about it I could feel the stress ulcerating the lining of my stomach, upholstering the walls of my coronary arteries, and overwhelming my immune system with an avalanche of corticosteroids. A ride like that would take years off my life. But my grandson was over the moon.

By what alchemy had the brains of the laughing riders transformed into exaltation an experience that in me would evoke only misery and fear? It's not that we understood the risks any differently, at least not cognitively. I know, after all, that most of the time nothing really bad happens in amusement parks. It's not that the external physical stressors were any less for them than they'd be for me, dangling head down ten stories above the ground. Styles of conscious stress management don't account for it either; there isn't time for those. The difference is in the ways our individual minds work to convert a concrete situation into an experience of either excitement or terror. Who is sane and who is crazy—the excited teen or the cautious grandfather?

The Grant Study's second lesson, and perhaps the one dearest to my heart, is that any exploration of the links between positive mental health and psychopathology requires an understanding of adaptive coping. As cough, pus, and pain remind us with disconcerting regularity, the processes of illness and the processes of healing look startlingly alike.

In this chapter, I will use the terms adaptation, resilience, coping, and defense interchangeably; likewise unconscious and involuntary. The study of psychological adaptive mechanisms began for me years before I became involved with the Grant Study. I had learned in an earlier longitudinal study to admire the means by which some people manage to achieve lasting remission from schizophrenia and heroin addiction. But I was not interested only in the resilience that comes with seeking social supports and devising ingenious conscious coping strategies. We've all got a few of those, and we know about them. I was interested in involuntary coping, analogous to the ways we clot our blood and send white cells out to fight infection. Or transform terror, like my grandson did on that roller coaster.

What makes the study of defenses so fascinating is the ambiguity of the boundary between psychopathology and adaptation. Early nineteenth-century medical phenomenologists viewed pus, fever, pain, and cough as evidence of disease, but less than a century later their colleagues had learned to recognize these "symptoms" as involuntary efforts of the body to cope with mechanical or infectious insult. Similarly, psychological defense mechanisms produce behaviors that may appear pathological to others (or even at times to us), but in fact reflect efforts of the brain to cope with sudden changes in its internal or external environment without too much anxiety and depression.

We depend physiologically on multiple elaborate systems of homeostasis, the task of which is to buffer sudden change; we don't faint when we stand up quickly, for example, because our bodies adjust our blood pressure to the demands of the new position. The task of the psychological homeostatic system that I am calling involuntary coping is to buffer sudden change in the four sources of mental conflict: relationships, emotions, conscience, and external reality. Defenses are extremely important to comfortable and effective functioning, like our other homeostatic systems. But they are difficult to study. They resemble hypnotic trance in that their use alters the perception of both internal and external

reality, and may compromise other aspects of cognition as well.

Even after I got to the Study and the investigation of unconscious coping became a central focus of its research activities, we never applied for an NIMH grant to inquire into this phenomenon. In fact, from 1970 to 2000 defenses took a backseat when we asked for money. Times had changed since the days when psychoanalysts were influencing the research agenda of the NIMH, and by 1970, defenses were too unfashionable for NIMH support. Yet at the Study, we were finding that defensive style predicted the future as well as, if not better than, any other variable we possessed.

The scientific community was trying to do its job when it dismissed defense mechanisms as a holdover from the (now outmoded) metaphysics of psychoanalysis. Yet defenses are real enough. They're elusive, yes, but not like fairies, yetis, and UFOs are elusive, and the other such will-o'-the-wisps that somehow always manage to elude our cameras. Defenses are more like rainbows and lightning and mirages—they're fleeting, but they can be photographed, replicated, and, above all, explained.

It was the many decades of detailed Grant Study recording, like sequential photographs or the transcripts of a neutral observer, that allowed us to identify in real behavior coping measures that are usually invisible to their users. This was necessary, because defenses are unconscious and involuntary. If I say, "You're projecting!" you will reply (probably angrily), "No, you're projecting!" and an argument is on that even an outsider wouldn't be able to resolve. A major contribution of the Grant Study, and one of its most appreciated results, has been to make the scientific study of defenses respectable.

WHAT ARE DEFENSES (INVOLUNTARY MENTAL COPING MECHANISMS)?

In 1856, Claude Bernard, a French physiologist and a founder of experimental medicine, started us on our way to understanding adaptation to stress when he wrote, "We shall never have a science of medicine as long as we separate the explanation of the pathological from the explanation of normal, vital phenomena." ¹ Pus, cough, and fever are certainly unpleasant, and sometimes dangerous. But they can also be lifesaving; it is these superficially pathological homeostatic responses to physiological stress that in many cases permit us to survive it. In 1925, Adolph Meyer, a founder of modern American psychiatry and an early consultant to the Grant Study, believed that there were no mental diseases, only characteristic reactions to stress. ² He thought that while patterns of mental reaction like denial, phobias, and even the projections of the paranoid character may look like illness, they may in fact be examples of Bernard's "normal, vital phenomena," promoting adaptation, healing, or at least psychological time-out. Just as fever, clotting, and inflammation use mechanisms that disrupt ordinary bodily equilibriums to do their healing work, so defense mechanisms heal through characteristic disruption of ordinary mental processes.

Figure 8.1 The four sources of intrapsychic conflict.

As outlined in Figure 8.1, defenses deflect or deny sudden increases in emotional or biological intensity, such as the heightened aggression and sexual awareness of adolescence. Psychoanalysts call this source of conflict id, fundamentalists call it sin, and cognitive psychologists call it hot cognition. Neuroanatomists locate it in the hypothalamic and limbic regions of the brain.

Defenses also enable individuals to mitigate sudden upsurges in guilt, such as might occur when a child puts a parent into a nursing home. Psychoanalysts call this source of conflict superego, anthropologists call it taboo, behaviorists call it conditioning, and the rest of us call it conscience. Neuroanatomists point to the frontal lobe and the amygdala. Conscience is not only the result of admonitions from our parents absorbed before age five, or even of cultural identifications; it is also formed by evolution, and sometimes by the irreversible learning that results from overwhelming trauma.

Defenses can moderate sudden conflicts with important people, living or dead, and protect us from the vulnerability and intensity aroused by sudden changes in intimacy. When a business partner walks out; when a marriage proposal is accepted; when a beloved child receives a fatal diagnosis—situations like these make for anxiety, excitement, or depression that can feel unbearable. In an adolescent striving for identity, even parental love that was once accepted without ambivalence may be unconsciously distorted for a while to make room for psychological separation to take place.

Finally, defenses allow us a period of respite, when necessary, to master inescapable realities that cannot be integrated immediately. They provide a mental time-out without which the individual would become acutely anxious and depressed. That is what would have happened to me had I been so imprudent as to join my grandson on the roller coaster. The events of 9/11 demonstrated this dynamic on a very large scale; the change in self-image resulting from an amputation is a smaller-scale example.

Over a period of forty years, Freud discovered most of the involuntary coping mechanisms that we recognize today, and identified five of their important properties. 3

- Defenses are a major means of mitigating the distressing effects of both intense emotion and cognitive dissonance.
- They are unconscious.
- They are discrete from one another.
- Although they may look like mental illness—sometimes like severe mental illness—defenses are dynamic and reversible.
- They are potentially as adaptive, even creative, as they are pathological.

I will add one last property to Freud's list:

- To the user, defenses are invisible; to the observer, defenses usually appear as odd behavior.

In 1971 the Grant Study offered a hierarchy of defenses from the psychotic to the sublime. 4 In 1977 and 1993, we were able to show, not just tell, how defenses worked. 5 Over the last twenty years, Cramer, and Skodol and Perry, have reviewed several empirical studies investigating the clinical value of retrieving defenses from Freudian oblivion. 6 As a result, the fourth diagnostic manual for the American Psychiatric Association (DSM-IV) finally organized defenses into a hierarchy of relative psychopathology similar to ours, and formally included it as an optional diagnostic axis. 7

A HIERARCHY OF DEFENSES

All defenses can effectively minimize the experience of conflict, stress, and change, but they differ greatly in their consequences for long-term psychosocial adaptation. Here they are organized into four levels, from least to most mature.

The psychotic defenses include delusional projection, psychotic denial, and psychotic distortion. They involve significant denial and distortion of external reality. They are common in young children and in dreamers, as well as in psychosis. To alter them requires altering the brain—either by maturation, by waking, or by the use of neuroleptic drugs.

The immature defenses include acting out, autistic fantasy, dissociation, hypochondriasis, passive aggression, and projection. Immature defenses externalize responsibility and are the building blocks of character disorders. They are familiar to most of us by observation. Immature defenses are like cigars in crowded elevators—they may feel innocent to the user, but observers often experience them as deliberately irritating and provocative. Defenses in this category rarely respond to verbal interpretation alone.

The intermediate defenses include displacement (kicking the dog instead of the boss), isolation of affect or intellectualization (separation of an idea from the emotions that go with it), reaction formation (turning the other cheek), and repression (keeping the affect visible but forgetting the idea that gave rise to it). Intermediate defenses keep potentially threatening ideas, feelings, memories, wishes, or fears out of awareness. They are frequently associated with anxiety disorders, but they are also part of the familiar psychopathology of everyday life, and they may be seen clinically in amnesias and in displacement phenomena like phobias, compulsions, obsessions, and somatizations. Intermediate defenses tend to be uncomfortable for their users, who may seek psychological help for that reason, and they respond more consistently than the lower-level defenses to psychotherapeutic interpretation. Intermediate defenses are common in everyone from the age of five until death.

The mature defenses include altruism, anticipation, humor, sublimation, and suppression. By allowing even anxiety-laden feelings and ideas to remain in awareness, they promote an optimum balance among conflicting motives and maximize the possibility of gratification in complicated situations. Altruism (doing as one would be done by), anticipation (keeping future pain in awareness), humor (managing not to take oneself too seriously), sublimation (finding

gratifying alternatives), suppression (keeping a stiff upper lip) are the very stuff of which positive mental health is made. Although they may appear to be under conscious control, unfortunately they cannot be achieved by willpower alone; just try to be really funny on demand. Furthermore, their deployment must be facilitated by others who provide empathy, safety, and example. If Gandhi had lived under Hitler instead of Churchill, he would have been a victim, not a hero.

Even mature defenses alter feelings, conscience, relationship, and reality in the service of adaptation. But they achieve their distortions gracefully and flexibly. Observers tend to regard the adaptive defenses as virtues and to experience them as empathic; their success depends on sensitive connections with others, as can easily be seen in humor and altruism. The immature defenses are not considered virtuous at all, and are usually experienced as manifestations of narcissism.

ASSESSMENT OF DEFENSIVE STYLE (IDENTIFICATION OF DEFENSES)

We can't see the spinach caught in our own teeth, and it's difficult to identify our own defenses. This means that self-report measures to assess defenses have limited validity. Even when a person consistently and correctly recognizes when he is being defensive, he may still misidentify the kind of defensive behavior he is using (that is, he may label it incorrectly). Furthermore, "defense mechanisms," like "character traits," are abstractions. What does it mean to distinguish between mature or empathic coping ("good" denial) and immature or narcissistic/unempathic coping ("bad" denial)?

An observer who wants to identify and appraise defenses objectively must triangulate between present behavior, subjective report, and past truth. Let's say, for example, that one woman founds a shelter for battered women and another breaks her toddler's arm in a tantrum. Defenses are unconscious; the first woman may attribute her altruistic behavior to a need to rent her house; the second may call the damage she did in her fury "an accident." It is only once we learn that social agency records from thirty years before reveal that both women had been taken at age two from the care of physically abusive alcoholic mothers that the defensive nature of both of their behaviors becomes apparent. Triangulation brings clarity to the defensive/adaptive nature of otherwise inexplicable behavior. You need the agency record and the mothers' explanations and the two tangible behaviors to label these unusual actions accurately.

The Grant Study's method of rating defenses can best be understood through a medical analogy. A symptom is a physical oddity. Perhaps the patient himself notices it, or perhaps a friend makes a comment. The patient may or may not understand what is going on. But when he goes to have it checked out, he'll be asked extensively about how it feels to him and what he notices about its context. He'll be examined; any relevant tests will be run. His self-report will be correlated with objective information from his medical history and recent workup. Eventually the symptom can be properly labeled and put in the context of its probable cause and mechanism.

Defensive behavior (including defensive symptoms or even creative products) is similarly called to our attention when something strikes us as odd or out of character. Once we've noted the oddity, we need to triangulate between what the subject says about it, what we know about his current circumstances, and the biological and biographical facts of his history. But objective documentation of mental health is scarcer and harder to come by than objective documentation of physical health. Like many other facets of mental health, therefore, the reliable identification of involuntary coping mechanisms requires longitudinal study. This is spelled out in much greater detail in my *Ego Mechanisms of Defense: A Guide for Clinicians and Researchers*. 8

In the Grant Study, we had thousands upon thousands of pages of the men's self-reports, and also a great deal of observational, historical, and objective material recorded by others. We triangulated among these to assess defensive style. When the men were 47, we picked out of their reports examples of behavior that we (or others) thought odd, selecting the ones that seemed most characteristic of the subject's way of operating. We considered them in conjunction with the observations of the many members of the Study staff over the decades, and also in the context of the historical information available to us, and the vast background of objective test and other documentary evidence that had accrued over the years. The selections that appeared on this scrutiny to be defensive were excerpted as 100-word vignettes. We compiled an average of two dozen such vignettes for each man. Then we asked blind raters to identify and label the vignettes for defensive style, as per the hierarchy above. We made certain to check the reliability of the raters—that is, to make sure that raters came to similar conclusions—and then we scored the resulting defenses for their level of maturity. One of Anna Freud's great contributions to the understanding of adaptation was her recognition that you can learn more about children's defensive styles by watching them play than by listening (as her father did) to free association and

dreams. And that’s essentially what we did; many of the College men enjoyed day jobs that were more like play than work.

There were two weaknesses in our exploration. We analyzed the defenses of only 200 of the men; this was an immensely time- and labor-consuming process, and we ran out of money before we could complete the work on the other 68. And our rater reliability was shaky on the defense of dissociation. For almost four-fifths of the men, however, and for all but one of the defenses, we had extensive and reliable identifications of characteristic defensive styles.

Table 8.1 illustrates the maturational levels reflected in our vignettes of adaptation at different ages. The Study found that adolescents were twice as likely to use immature mechanisms as mature ones. Young adults were more than twice as likely to use mature mechanisms as immature ones. Between ages thirty-five and fifty, individual men were four times likelier to choose mature mechanisms than they had been when they were adolescents.

Table 8.1 Maturation of Defenses Over the Lifespan

* The vignettes are the ministories excerpted from the men’s histories as examples of adaptive coping behavior, and then identified and rated.

We found that defensive style was relatively independent of nurture, and that was not a surprise. Almost by definition, adaptive mechanisms are the ego’s antidotes to a poor environment; its adaptive capacity reflects the difficulties it has mastered at least as well as the assets with which it was originally blessed. Like Godfrey Camille, some individuals became progressively better adapted as they mature. Folk wisdom recognizes this: Pearls are what oysters do about irritation, and what doesn’t kill you makes you stronger. Damaged children in their growing can find ingenious ways to compensate for earlier deprivation. Northwestern University psychology professor Dan McAdams recognizes this reality as the “redemptive self.” 9

Maturity of defensive style was not associated with parental education or social class, and bore no relationship to tested intelligence, body build, or physical responsiveness to stress. The use of mature mechanisms correlated very highly with variables that reflect psychiatric health and warm human relations. 10 And we found a very significant association between immature defenses and genetic vulnerability (that is, number of ancestors with major depression, alcoholism, or shortened longevity). The depressed and the alcoholic were most likely to use immature mechanisms later in life. Fifty percent of the ten men who had two mentally ill parents manifested immature patterns of defense, and only 9 percent of fifty-five men without mentally ill parents showed a predominantly immature pattern of defense. Of course, this finding could also be explained as identification (monkey see, monkey do).

DOES DEFENSIVE STYLE MATTER?

Once we had learned to rate maturity of defenses reliably, the next question was: Did choice of mechanism matter? Would the men’s coping styles tell us any more than their handwriting had about their lives? To answer this question, for each of three cohorts, the 18 frequently used individual defenses were clustered into the four groups (psychotic, immature, intermediate, mature) I’ve outlined above. Correlations were calculated between the Study member’s maturity of defenses and other indications of successful adult development. As Table 8.2 indicates, the associations were impressive. Maturity of defenses as measured from age twenty to forty-seven predicted the men’s Decathlon scores at age seventy to eighty with a correlation of .43, which is very, very significant. Choice of involuntary defense mechanisms clearly does matter. Furthermore, the positive associations between maturity of defenses and mental health were, like our other maturity ratings, independent of the social class, education, and gender of the Study members.

Let me depart from abstraction now to show what the hierarchy of defensive styles looks like in real life. Here are the life stories of two men who have also appeared in *Adaptation to Life*. A few words first, though, about the chief adaptive mechanisms of the men you’ll meet here.

Table 8.2 Correlations Between Maturity of Defenses and Measures of Successful Adult Outcome

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

n.a . = data not available.

Sublimation, suppression, intellectualization, and repression are four distinct ways of dealing with desires and impulses that are experienced as unacceptable or dangerous. In sublimation, the emotional energy fueling these desires is invested in other goals that are acceptable (and frequently also socially valuable). In suppression (sometimes called stoicism), the desires and emotional energy are held together in consciousness, and the anxiety and depression they evoke either tolerated or ignored, until a way to pursue them safely and appropriately is discovered. In intellectualization (sometimes called isolation of affect), forbidden ideas remain in awareness but divorced from their emotional intensity. In repression, unacceptable desires and impulses are excluded from awareness along with the ideation associated with them, but the emotion remains powerfully present, with no provision made for immediate or future gratification. Like all of the defense mechanisms I'll be discussing in this chapter, these four are not employed deliberately or even consciously; their use is automatic and involuntary, like the mobilization of fever in the face of infection.

Repression is a short-term solution to the complexities and contradictions of emotional life. Our desires and impulses often have to be handled with care, but to dismiss them wholesale from awareness is to dismiss a great deal of emotional vitality as well. Repression serves us best, therefore, as a respite from conflict and confusion, a temporary battening down of the hatches until we are ready to look for more enduring ways to keep from being overwhelmed by our feelings. Intellectualization is much the same. It permits forbidden thoughts and desires to remain in awareness, but without their passionate emotional valence. Intellectualization allows for cerebral satisfaction and sometimes very valuable advances in understanding—one Grant Study man grew his mother's cancer in his lab as a tissue culture—but at the risk of alienation from emotional experience.

Sublimation and suppression are two more satisfactory long-term adaptive techniques. Sublimation—finding outlets for our passions—is the gift that allows successful artists to work themselves into the very marrow of our bones. In general we do not like being made to weep, but when Mahler, Verdi, Plath, Shakespeare, and Jesus do this, we bless them for transmuting the bitter poison of their own lives into an elixir of salvation.

In comparison, suppression is about as glamorous as a Mack truck. Perhaps less mysterious than the other three, it is an important workhorse of an adaptation, and a very successful one. Of all the defenses it was the one most associated with being well integrated in college, and with excellence at the events of the Decathlon three decades later. Sublimation, for all its elegance, was less associated with success or with joy. Artists are not known for freedom from mental illness, as Beethoven exemplifies—only for the joy they bring to others.

One way of coping with the hurly-burly of reality is to retreat into the pleasures of the mind. Several of the Grant Study men who went into teaching used this type of sublimation as a successful coping style. They tended to have productive and distinguished careers, like Daniel Garrick and Dylan Bright (below). The professors like Peter Penn who relied heavily on intellectualization as their major coping mechanism, however, tended to experience their jobs and their marriages as sterile, and they showed far more evidence of emotional problems.

THE LIFE OF DYLAN BRIGHT

Professor Dylan Bright was a vivid illustration of the coping potential of sublimation. Although he was less gifted intellectually than the average Study member, an exciting luminescence surrounded his life. As soon as I walked into his office, he put his feet up onto the desk and started to talk. He looked more like a prizefighter than an English professor, and I was moved by his affective richness. But initially I was not quite sure that I liked him. His first response to my request for an interview was, "Christ, that kills the afternoon!" His aggression was barely tamed and verged on the abrasive, and only his charm kept me from regarding the encounter as a pitched battle. He graphically described his worries and then growled, "If they get out, I'll kick your teeth in."

Bright was a football linesman and champion wrestler who, almost as an afterthought, became a professor of poetry. In high school he greatly preferred the excitement of athletics to the dreary world of the classroom; he was a rebellious D student, and at one point was nearly expelled. Nevertheless, his headmaster saw Bright as "vibrant and ardent in his beliefs," and the Study staff perceived him as "an eager, enthusiastic, attractive youngster with an outgoing personality." Bright's intensely competitive spirit, which he eventually harnessed in the service of his academic career—that is, sublimated—was never extinguished. His energy, his extraordinary capacity to win close friends, and his knack for exciting hedonistic activities made him one of the most dramatic members of the Study.

Unlike suppression and anticipation, the use of sublimation was not associated with particularly happy childhoods. Like Beethoven, Bright grew up in a family filled with turmoil. His father was an emotionally unstable alcoholic who was rarely at home and whose hobby was hunting. Dylan saw his parents' marriage destroyed by fighting, and early on he tasted both the triumphs and the dangers of (figuratively) taking the place of his father. His mother was an exuberant and energetic woman who was three inches taller than her son even after he reached adulthood. Her charm was appreciated by several Study observers, but from the beginning she taught Dylan to beware of instinctual pleasure. Before his first birthday he had been cured of thumb-sucking, bedwetting, and soiling himself. At two, his mother made him wear mittens to bed because of his "perfectly revolting" habit of masturbating. As a child, he conceived of God as "a person looking down on me, ready to conk me on the head with a thunderbolt."

In his lifelong quest to conquer fear, when Dylan Bright let go of his mother's skirts he became a daredevil quasi-delinquent. As a youth he incurred more cerebral concussions than anybody else in the Study—there's a statistic for you! But with the passage of time, his mastery became more graceful. After eighteen, he began concentrating on learning to do what he called "responsibly adventurous things," and there were no further injuries. First he was an all-state football linesman in high school, and then a fiercely competitive college wrestler. He played tennis for blood, shunning doubles for the joys of single combat. After college, once his devotion to tennis and wrestling had been replaced by an equally fierce devotion to poetry, he was still out to win. He raced through graduate school at Yale with top grades. He accepted an appointment at Princeton for the prestige, and a few years later exulted in his early acquisition of academic tenure.

Bright did not start out with creative and empathic ways of dealing with his feelings. As he and his ego matured, however, he replaced acting out (his delinquent rebellion) with reaction formation (containing emotional impulses by doing their opposite). For example, he suddenly found the first girl with whom he had slept "revolting"—unconsciously choosing the same word that his mother had used to condemn his sexual experimentation in infancy—and gave up intercourse with his next girlfriend out of an ascetic desire to "see if I could." In college, the once delinquent Bright seriously considered a career in law enforcement. As a young and lusty English instructor at Princeton, his devotion to enforcing parietal rules surprised the administration as much as it irritated the students. Even in middle life, Professor Bright conceived of his success in terms of rigid control. "If a person does not have self-discipline," he cautioned me, "he can go to rot so fast."

Bright didn't consider the possibility that his rigid self-control might not in fact have been protecting him, or that it might even have been steering him toward an empty disaster of a life. But it was only as the reaction formation of his youth gave way to sublimation that he caught fire. He risked his amateur standing by wrestling in exhibition matches, but "consecrated" his illegal fees by investing them in violin lessons. During his nineteen-year-old abstention from sexual intercourse, he substituted a close and exciting intellectual friendship and made his first discovery of poetry. He fought to be first in his graduate school class, but he gentled his naked ambition by writing his Ph.D. thesis on the poetry of Shelley.

When Bright was thirty-five, his wife broke up what had been a very close marriage. At the same time, he was becoming aware that his scholarship, although adequate to win him tenure at Princeton, would never win him national recognition. This was a critical period. Faced by two real defeats, for a while he lost himself in alcohol, careless affairs, and stock car racing—the poetry professor regressed to adolescent acting out. But he quickly replaced his temporary delinquencies with more acceptable and productive quests for excitement. He went scuba diving in the Aegean with a close friend, and made a discovery that allowed him to reinterpret a line of Homer's *Odyssey*. "Oh," he told me, "that was a heady experience."

Bright's adaptive responses were ingenious indeed. Sexual abstinence (this time following upon his divorce) once again brought him into relationship with a brilliant colleague who remained a friend for life. He withdrew from academic competition in which he anticipated only defeat, and involved himself instead in activities that permitted him to master danger with minimum risk and at the same time to anesthetize grief with real excitement. Sublimation not only facilitated the efficient expression of his instincts, it also permitted Bright to avoid the labels of "neurotic" and "mentally ill." He had once described himself as "a laughing man. I just let things slip off my back." But he didn't do that by permanently drowning his troubles in alcohol, or in self-destructive chance-taking, or in Scarlett O'Hara's ambiguous mantra of denial, "I'll think about it tomorrow." His capacity for sublimation enabled him to change the terms of his life. He went

on the wagon to control his incipient alcoholism. He made a successful second marriage. He remained in touch with his feelings while softening them with excitement, laughter, and people. Asked if he had ever seen a psychiatrist, he spoke instead of his second wife and his best friend: "Professional assistance would be a pale shadow compared to these companions." Like art, love is an act of creation, but love far surpasses art as a cure for emotional suffering. Bright died at sixty-two. Eighteen years of heavy drinking and forty-five years of a two-pack-a-day habit had led to lung cancer. But like the Marlboro man, he died with his boots on.

THE LIFE OF FRANCIS DEMILLE

Francis DeMille grew up in suburban Hartford. He never knew his father, a businessman who left home before his birth and died shortly thereafter. His father's relatives played no part in his upbringing, and the DeMille household consisted only of Francis, his mother, and two maiden aunts. From one to ten, he lived in an entirely female ménage, with a playroom that he used as a theater. He was encouraged to play by himself on this stage; in fact his mother proudly reported to the Study that he "never played with other boys."

When he joined the Grant Study in 1940, Francis DeMille looked hardly old enough to be in college. His complexion was as fresh as a girl's, and despite his erect carriage, he struck several observers as rather effeminate. He was in the top 8 percent in "feminine" body build. But he also impressed the staff with his charm. His manner was open, winning, and direct, and he discussed his interest in the theater with a cultivated animation.

The staff psychiatrist marveled that at nineteen DeMille had "not yet begun to think of sexual experience." In fact, Francis as a college student "forgot" to an astonishing degree to think about sexual fantasy, aggressive impulses, or independence from his mother. He didn't remember his dreams well either, and he reported that "distressing emotional reactions fade quickly." He didn't date in college, he totally denied sexual tension, and he blandly observed, "I am anything but aggressive." He was a poster child for repression.

In retrospect it's a little hard to understand how DeMille came to be included in a study of normal development. It seems that dramatic skill got him in. The staff may have wondered at his sexual oblivion, but still they perceived him as "colorful, dynamic, amiable, and adjusted." Francis took an active and enjoyable part in college dramatics; the Study staff thought that his mother was pushing him into the theater, but Francis seemed unaware of that. Characteristic of people who use repression as a major defense, he reported that he preferred "emotional thinking to rational thought."

Like many actors, DeMille was also a master of dissociation, or neurotic denial. He found it "revitalizing" to free himself from inhibitions by becoming someone else in a play, and so "venting my emotions." Despite the fact that the staff worried about his inner unhappiness, during psychiatric interviews he seemed "constantly imbued with a cheerful affect." What, me worry? Alfred E. Neumann was good at dissociation, too.

When Lieutenant DeMille managed to remain at his mother's side both emotionally and geographically through the whole of World War II, the Study internist began to fear that he was going to be a lifelong neurotic. The Navy never took him farther from Hartford than the submarine base at Groton, Connecticut. But it was in the Navy, with continued maturation, that DeMille's repression at last began to fail. At first this was disconcerting and quite anxiety-provoking. He became aware of his peculiar lack of sexual interest, and was fearful about possible homosexuality. Discussing this problem in a Study questionnaire, he made, as many repressed people do, a revealing slip of the pen. "I don't know whether homosexuality is psychological or psychiological in origin." As it turned out, his unconscious was right; there was nothing physiologically wrong with his masculinity. Later in life he would father three children.

In manageable doses, anxiety promotes maturation, and over time DeMille began to replace repression and dissociation with sublimation. He wrote to the Study that he was always rebelling against the Navy, standing up for his own individuality and for that of his men. His own reports of his behavior might have led us to call this passive aggression, an immature and destructive defense. But his military efficiency records gave him his highest officer efficiency rating in "moral courage" and "cooperation." This heretofore compliant man had turned at least one rebellion into a veritable work of art that even the military could appreciate.

By twenty-seven, DeMille's worries about possible homosexuality had been put to rest. "I enjoy working with girls!" he announced joyfully. He had found a job teaching dramatics at Vassar. He managed the short move from Hartford to Poughkeepsie, thus gratifying the "great necessity I feel for breaking away from home." Three years later, he shattered

maternal domination still further by marrying an actress whom he had once directed in his amateur theatrical group. His marriage today may not be the best in the Study, but it is above average and it has survived for more than fifty years.

DeMille was also becoming more aware of his use of repression as a defense. In reply to a questionnaire that asked him about his marital sexual adjustment, he wrote, "I must have a mental block on the questionnaire. My reluctance to return it seems to be much more than ordinary procrastination." He was in conflict over his sexual adjustment, and he knew it. He was in conflict over work, too, mentioning in that same questionnaire some concerns about the desire to work aggressively for money. He could recognize now that on the one hand he needed to feel that he wasn't greedily seeking money, but that on the other, "I didn't tackle the career problem properly." However, once involuntary coping styles have become conscious, they no longer "work." And with insight comes resistance. That was the last the Study was to hear from DeMille for seven years. He never saw a psychiatrist during this time. But in keeping with his new capacity for sublimation, he wrote a successful comedy for his amateur theatrical group entitled *Help Me, Carl Jung, I Am Drowning*. And after he returned to the Study, he revealed the following ingenious way in which his ego had sublimated his issues with money and aggression.

If DeMille was more mercenary than he had wanted to think, his solution to the conflict was artistic. In his twenties, he had vowed that he would never associate himself with the "specter of American business," but in fact he left Vassar to return to his mother's city of Hartford, where Insurance is king. There, despite his theatrical interests, he became a very special corporate success story. In an industry not known for its openness to individual expression, he crafted a niche on the advertising side that gave him autonomy, high management perks, and a chance to exercise his dramatic flair. He took pains, however, to assure me that his success in the marketplace threatened no one and that he was not "overly aggressive." As he put it, "The ability to stay alive in a large corporation took all the craftiness I have." Only in his community theater group could he unashamedly enjoy playing aggressive roles.

In an interview at age 46, DeMille shared a vivid new recollection of a hyper-masculine uncle who had been an important, if previously unmentioned, role model during his adolescence. This was the recovery of a lost love, and another softening of his stern earlier repression of masculine role models, and all without psychotherapy. Five years after our interview, DeMille elaborated further on this uncle, whom he called "the only consistent male influence—very dominant—a male figure that earlier I had rejected." But not entirely, it seems, for with his pipe, his tweed jacket, his leather study furniture, and the bulldog at his feet, the middle-aged DeMille now rather resembled his uncle. The charming emotional outpourings of his adolescence were gone; he now hid emotions behind lists, order, and a gruff, hyper-masculine exterior. "In college," he said, "I was in a Bohemian fringe; but I've changed since twenty-five years ago. Maybe some clockwork ticked inside me and made me go down this route." Perhaps this new identification explains why a few years earlier he had given up his mother's religion and was "suddenly smitten" with the Episcopalian tenets of the father he had never known. Certainly he hoped that his sons would never discover that their father had once worn long hair; by 1970 DeMille thoroughly disapproved of unshorn locks. Folks do change.

At sixty, after his mother died, DeMille took early retirement, delighted to be liberated from corporate life. But before he left his Hartford suburb for rural Vermont, he was honored as "first citizen" of the town to which he had given sixteen years of devotion, finally including a stint as chairman of the town historical board.

Once he was settled in Vermont, an acknowledged Guardian, his community service only increased. Over fifteen years of retirement he led one successful fundraiser after another. He rebuilt the small community church; he made possible the new library. He wrote, directed, and starred in plays for the town summer theater. He coped with life's kicks in the teeth with composure and admitted that he wasn't too much of a patriarch to let his wife take care of all the taxes and bills. But he still said of himself that "my mind is blank to things I don't want to remember."

"Sometimes make-believe and reality get mixed up," he had confessed at forty-seven. DeMille was a man who had always situated himself in places where he had access to a stage and did not have to play exactly by the rules. As a child he had preferred his playroom to the schoolyard; at the insurance agency he constructed for himself a special niche where he could do exactly what he wanted. Still, he had built a library and a church where neither had existed before. And in three different communities he was the town historian whose remembering of the past—admittedly, someone else's past—served the future. And in retirement his devotion to play and to plays was no longer a problem; in fact he was rewarded for it. He did not look for tragic roles like Lear; instead, he adapted realistically by playing (for pocket change)

the leading role in *On Golden Pond*. As playwright and director, he could make life just the way he wanted it, and the results were real, not make believe. For Garrick, acting was a living and a passion; for DeMille it was a coping mechanism.

DeMille didn't begin "paring down" until after he turned seventy-five, when he was still taking seven-mile hikes and was still a community star. Until eighty, he saw life as a "pretty good ride," but after that he began to fail. He cut his walks back to just a mile. His wife died when he was eighty-five, and he developed dementia. He survived to ninety, but in a nursing home and unable to walk. Mature defenses affect how we feel about aging, but they can't, alas, guarantee that we'll live happily to a hundred and then collapse all at once like the one-hoss shay. It took the Grant Study twenty-five years to teach me that Daniel Garrick had been very lucky indeed.

PROSPECTIVE VALIDATION OF THE HIERARCHY

Do mature defenses make it easier for people to find joy in living? Or is it that joy in living allows us the luxury of mature defenses? I wanted to know whether maturity of defensive style has predictive validity as well as the uniformly positive correlations in Table 8.2. Predictive validity means that an association is not just a statistically significant coincidence, but that it can—as the term implies—predict the future reliably.

We approached the question this way. Raters unfamiliar with the lives of the College men before they turned fifty assessed their joy in working, their use of psychiatrists and tranquilizers between the ages of fifty and sixty-five, the stability of their marriages, and the course of their careers (that is, whether they had progressed or declined) since age forty-seven.¹¹ This assessment was then correlated with an assessment of defensive style. Mature defensive styles assessed between ages twenty and forty-seven were very significantly predictive of a superior adjustment at sixty-five for the College men (see Table 8.3). Only two men of the thirty in the bottom quartile of defenses (as assessed between twenty and forty-seven) were in the top quartile of adjustment at age sixty-five, and the mean Decathlon score for these thirty men was only 1.4 at age eighty. The men with the most mature defenses had Decathlon scores three times higher (4.6), a very significant difference, and of those thirty-seven men only one was in the bottom quartile of adjustment at sixty-five.

Then we contrasted the defenses of the twenty-three College men who at some point in their adult lives were clinically depressed with those of the seventy least distressed College men (that is, the men who over thirty years of observation eschewed both tranquilizers and psychiatrists, and never appeared to merit a psychiatric diagnosis). The two groups showed a very significant difference in the overall maturity of their defenses.

Table 8.3 Late Life Consequences of Mature Defenses at Age 20–47 for the College Cohort

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

* Sample size reflects the fact that we did not have information on all the men on all the variables.

Table 8.4 Use of Defenses by the Most Depressed and Least Distressed Men in the College Sample

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

As shown in Table 8.4, 61 percent of the least distressed and only 9 percent of the most depressed men exhibited generally mature defenses; whereas 53 percent of the depressed men and only 9 percent of the least distressed men consistently favored less mature defensive styles. Although altruism, a mature defense, is used by many people with unhappy childhoods to master adult life, sustained use of altruism was never noted as a major adaptive style among the depressed College men.¹² Instead, the most depressed men were more likely to use reaction formation and passive aggression, whether anger was directed against others or themselves.

Here, too, simple association does not prove cause. Immature defenses are associated with alcohol abuse and brain damage, but they do not cause them; rather, both alcohol abuse and brain damage can cause regression in maturity of defenses. Similarly, the association of immature defenses with depression is likely not a simple one. In some people, severe depression probably leads to a regression in adaptive style as less mature defenses, once developmentally surpassed, come back to the fore. In others, depression and immature defenses may both be responses to unmanageable stress, disordered brain chemistry, or both. Immature defenses probably predispose some people to depression. Much

more evidence is needed to clarify the relationship between affective disorder and maturity of defenses.

To me, perhaps the most fascinating question of all was: Does the assessment of defenses in young adulthood predict future physical health? For the first twenty years that I worked with the Study, I fervently believed that because defenses mitigate stress, mature defenses would lead to better physical health than immature ones. In this belief I was wrong. For at least ten years after the men's defensive styles were recorded and rated at age forty-seven, the health of the men with mature defenses deteriorated less quickly. And, as I established in Chapter 7, immature defenses were one of the mental health variables that predicted decline in physical health in the period between forty and fifty-five. By age sixty-five, however, the association of mature defenses with continued good health could no longer be discerned. Once again, prolonged follow-up demolished both theory-based conviction and short-term evidence.

DEFENSIVE STYLE, GENDER, EDUCATION, AND PRIVILEGE

A major effort of the Grant Study on my watch has been to dissect the psychological mechanisms of homeostasis by which human beings achieve resilience in the face of sociocultural challenges. In biological medicine the task is easier. Blood-clotting is an elegant example of unconscious homeostasis, but the fact that the Romanovs died young from hemophilia and their peasants did not was not a class issue; on the contrary, it was a trumping of social class by genes. Clotting factors are distributed in an egalitarian fashion. It would be nice to think that biology is as democratic in distributing coping skills as it is about clotting factors and immune mechanisms, but there is room for doubt. Many aspects of mental health are a function of education, IQ, social class, and/or societal gender bias.

It is of great interest, then, that maturity of defensive style did not seem to be affected by socioeconomic status, intellectual ability, or gender. It is true that the College sample looked a little better than the Inner City men in this regard (11 percent and 25 percent respectively used predominantly immature defenses). But that can be explained by the original selection process; the College men were selected for mental health, while the Inner City men were not; in fact they had been deliberately matched with delinquents. Mental health and defenses are closely correlated.

A less distorted view of the effect of privilege on defensive style can be achieved through within-group comparison—that is, by comparing the members of one group to each other. In this way initial selection bias is circumvented. Table 8.5 examines the effect of social class, IQ, and education upon differences in defensive maturity within three different groups. The associations are insignificant. Even the relationship of a warm childhood environment to maturity of defense is less than one might expect.

Table 8.5 Correlations Between Maturity of Defenses and Biopsychosocial Antecedents

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

* Thirty men with IQs less than 80 were excluded.

The effect of cultural diversity was tested in the following manner. Sixty-one percent of the Inner City men had parents who had been born in a foreign country, but the men themselves had all grown up in Boston, were fluent in English, and had been sampled and studied in the same way. Thus it was possible to vary ethnicity and culture of rearing while holding other demographic variables constant. In some facets of adult life, parental ethnic differences among the Inner City men were seen to exert a profound effect. For example, as I'll discuss in Chapter 9, men of white Anglo-Saxon Protestant (WASP) and Irish extractions had rates of alcohol abuse five times those of men of Italian extraction. But there was little cultural difference in defensive style. Dissociation was the only defensive style that appeared to be used significantly more by the WASPs than by Italians. 13 (Dissociation was also the defense with the lowest rater reliability.)

But if culture appears to have little effect on defensive style, this is not true of biology. The central nervous systems of some of the Inner City men had been impaired by chronic alcoholism. (By this I do not mean acute intoxication; most men were quite sober when interviewed.) In addition, some men had possible early cognitive impairment, as suggested by IQs less than 80. Both these groups exhibited significantly less mature defensive styles than the rest of the Inner City men. All immature defenses were two to four times as common in these two compromised groups as in the unimpaired sample.

CONCLUSION

Empirical investigation provided clear answers to the Study's three major questions about involuntary coping mechanisms. First, maturity of defenses can be rated reliably. Second, maturity of defenses demonstrated predictive validity toward future mental health. Third, maturity of defenses is independent of social class and gender, but is affected by biology.

Defense mechanisms are not just one more dogma of the psychoanalytic religion. On the contrary, the brain's mechanisms of involuntary adaptation are a fit subject for serious study by social and neurological scientists. But, as with all matters of lifetime development, long and deep access is needed to study them. It was the unusual longitudinal and naturalistic nature of the Grant Study that permitted the conclusions in this chapter. Fledgling efforts are now under way to image what the brain does while deploying defenses, but scientists will likely have to wait upon advances in brain imaging technology for further confirmation. 14